

Strict Class 1 Reservation vs FCFS

Utilization, Service-Rate, and Waiting-Time Tradeoffs

Short preliminary decision memo

Executive Summary

Bottom line. These results are preliminary and use only 5 seeds. Near the current baseline ($\lambda_1 = \lambda_2 = 25$, total $\lambda = 50$, $S = 32$, $H = 14$), strict reservation does **not** beat pooled FCFS once the service-rate guardrail is checked. Both objective studies select $Q = 0$ at the baseline.

- **Study A, utilization-first:** choose Q by booked-slot utilization, $U_{\text{util}} = \rho^{\text{booked}}$. At the baseline, the best choice is $Q = 0$ and the strict-minus-FCFS difference is $+0.000$.
- **Study B, wait-adjusted:** choose Q by booked-slot utility with Class 1 weight, a protected-slot penalty, and an offered-wait penalty. With $w_1 = 1.5$, $w_2 = 1.0$, $c = 0.05$, and $\gamma = 0.05$, the baseline best choice is also $Q = 0$.
- **Guardrails matter.** Several high- Q settings look better under the wait-adjusted utility, but they do so by shifting service strongly toward Class 1. Those settings fail the guardrail $\min_i \text{served_rate}_i \geq 0.55$ and should not be treated as usable wins.
- **Interpretation.** Study A is the safer first-pass rule because it directly asks whether appointment capacity is booked. Study B is useful for understanding priority and wait tradeoffs, but only after service-rate constraints are enforced.

1 Policy And Question

The simulation compares two booking rules:

Pooled FCFS

All patient classes compete for the same daily appointment capacity.

Strict Class 1 reservation

Q slots per day are protected for Class 1. Class 1 can use protected slots first and then general slots. Class 2 can only use general slots. Unused protected slots remain empty.

The objective functions below are **after-the-fact scoring rules**. They do not change the simulation policy. They are used to compare FCFS and strict reservation under matched scenarios and matched seeds.

The main question is:

When, if ever, does strict Class 1 reservation outperform pooled FCFS, and what value of Q is best once utilization, service rates, waiting time, priority, and loss of pooling flexibility are included?

2 Notation And Baseline

The notation follows the TeX reference files used by the simulation:

Symbol	Meaning
S	daily appointment capacity. In this run, $S = 32$.
H	rolling booking horizon. In this run, $H = 14$.
Q	protected Class 1 capacity per day. This is the policy parameter.
λ_i	class- i daily arrival rate. Baseline: $\lambda_1 = \lambda_2 = 25$.
λ	total daily arrival rate, $\lambda = \lambda_1 + \lambda_2$. Baseline: $\lambda = 50$.
τ	offered delay, in days.
ρ^{booked}	booked-slot utilization: booked service-day slots divided by available measured slots.
ρ^{attended}	attended utilization: completed visits divided by available measured slots.
served_rate_i	Class i served rate: served Class i arrivals divided by Class i arrivals.

The demand sweep used here has equal arrivals for the two classes:

$$\lambda_1 = \lambda_2 \in \{20, 25, 35, 50\},$$

so total demand is $\lambda \in \{40, 50, 70, 100\}$. The seed set is 5101–5105.

3 Objective Functions

3.1 Study A: Utilization-First With Service-Rate Guardrails

The main score is booked-slot utilization:

$$U_{\text{util}}(Q) = \rho^{\text{booked}} = \frac{\text{booked service-day slots}}{\text{available measured slots}}.$$

Booked-slot utilization is used because it captures whether appointment capacity was scheduled, including slots that later became no-shows. Attended utilization remains a diagnostic because it measures completed visits.

The service-rate guardrail is:

$$\min\{\text{served_rate}_1, \text{served_rate}_2\} \geq 0.55, \quad \text{served_rate}_i = \frac{\text{served}_i}{\text{arrivals}_i}.$$

The guardrail is not added to the objective. It decides whether a chosen Q is acceptable.

3.2 Study B: Wait-Adjusted Utility With Service-Rate Guardrails

The weighted booked-slot term is:

$$U_{\text{weighted_booked}}(w_1, w_2) = w_1 \rho_1^{\text{booked}} + w_2 \rho_2^{\text{booked}}.$$

The weighted offered delay is:

$$\bar{\tau}_w = \frac{w_1 \sum \tau_{\text{offered},1} + w_2 \sum \tau_{\text{offered},2}}{w_1 \text{ offered}_1 + w_2 \text{ offered}_2}.$$

The wait-adjusted objective is:

$$U_{\text{wait_util}}(Q) = U_{\text{weighted_booked}}(w_1, w_2) - c \frac{Q}{S} - \gamma \frac{\bar{\tau}_w}{H}.$$

For the headline comparison:

$$w_1 = 1.5, \quad w_2 = 1.0, \quad c = 0.05, \quad \gamma = 0.05, \quad \min_i \text{ served_rate}_i \geq 0.55.$$

The terms have simple roles: booked capacity is rewarded, Class 1 can be weighted more heavily, protected capacity is penalized, and longer offered waits are penalized after normalizing by the booking horizon.

4 Main Results

4.1 Baseline Comparison

At the baseline, both objective studies select $Q = 0$. The smallest positive strict-reservation value shown here, $Q = 4$, has almost identical booked and attended utilization but a lower Study B score after the protected-slot penalty.

Policy	Q	ρ^{booked}	ρ^{attended}	SR ₁	SR ₂	Min SR	Wait	Study A	Study B	Feasible
Pooled FCFS	0	1.000	0.912	0.586	0.583	0.582	5.419	1.000	1.233	yes
Selected $Q = 0$	0	1.000	0.912	0.586	0.583	0.582	5.419	1.000	1.233	yes
Strict, smallest positive Q	4	1.000	0.912	0.587	0.582	0.582	5.419	1.000	1.227	yes

4.2 Best Q By Scenario

The two studies agree in 4 of the 12 scenario rows. Where they disagree, Study B often selects a larger Q because it rewards Class 1 access and wait improvements. Those selections usually fail the minimum class served-rate guardrail.

Scenario	A: best Q	A feasible	B: best Q	B feasible	Agree
Symmetric, $\lambda = 40$	0	yes	0	yes	yes
Symmetric, $\lambda = 50$	0	yes	0	yes	yes
Symmetric, $\lambda = 70$	0	no	32	no	no
Symmetric, $\lambda = 100$	0	no	32	no	no
Class 1 advantaged, $\lambda = 40$	0	yes	0	yes	yes
Class 1 advantaged, $\lambda = 50$	0	no	24	no	no
Class 1 advantaged, $\lambda = 70$	0	no	32	no	no
Class 1 advantaged, $\lambda = 100$	0	no	32	no	no
Class 1 disadvantaged, $\lambda = 40$	0	yes	16	yes	no
Class 1 disadvantaged, $\lambda = 50$	16	yes	16	yes	yes
Class 1 disadvantaged, $\lambda = 70$	0	no	32	no	no
Class 1 disadvantaged, $\lambda = 100$	0	no	32	no	no

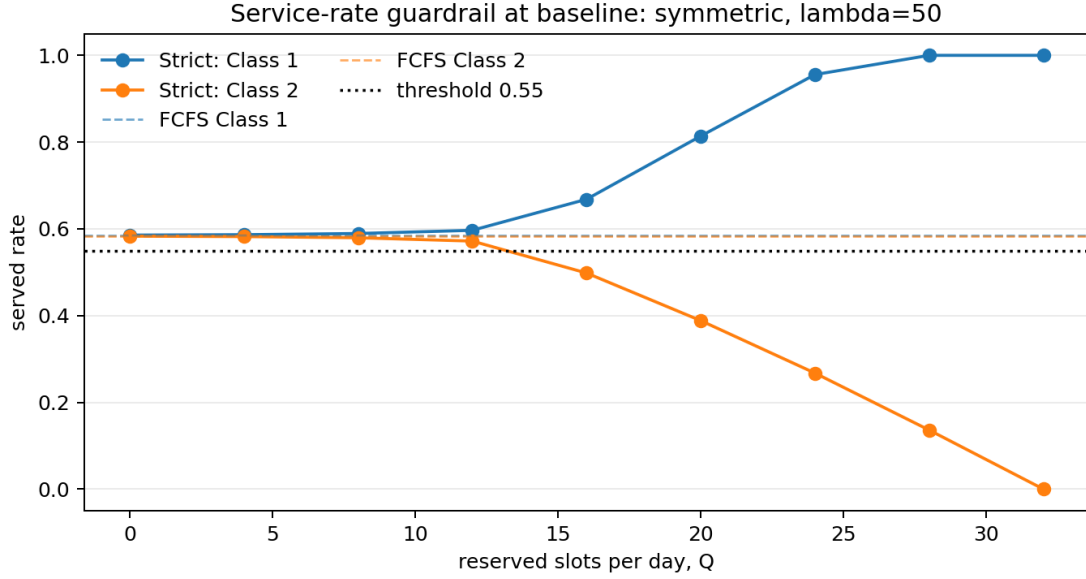


Figure 1: Baseline service-rate diagnostic. Dashed lines are FCFS references. The dotted line is the headline service-rate threshold, 0.55.

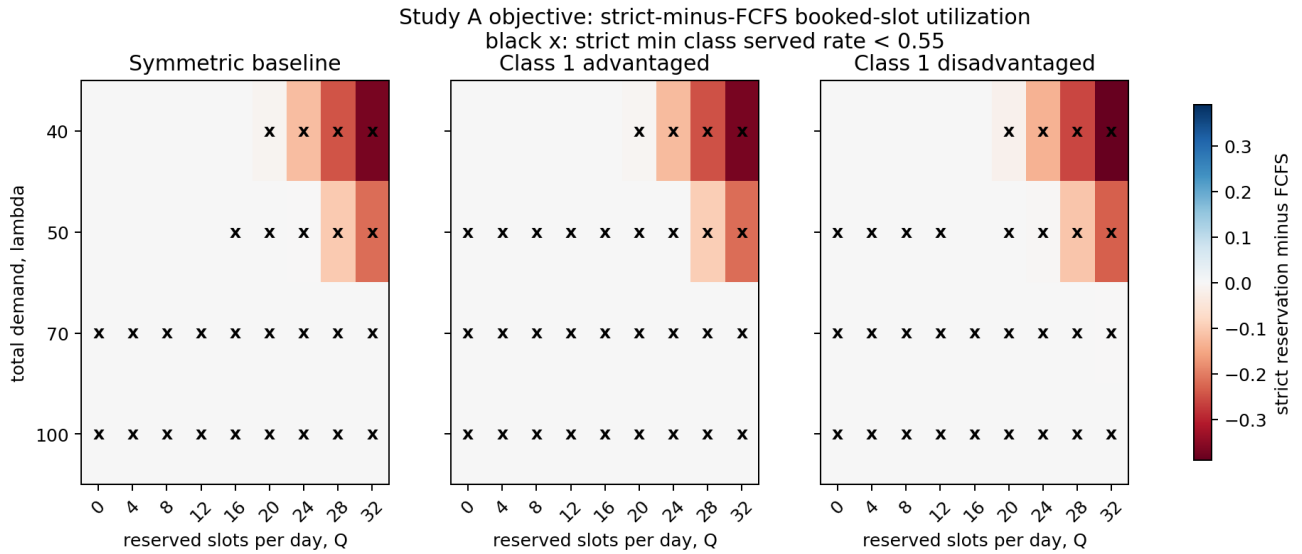


Figure 2: Study A objective: strict-minus-FCFS booked-slot utilization. Black x marks cells where strict reservation violates the service-rate threshold.

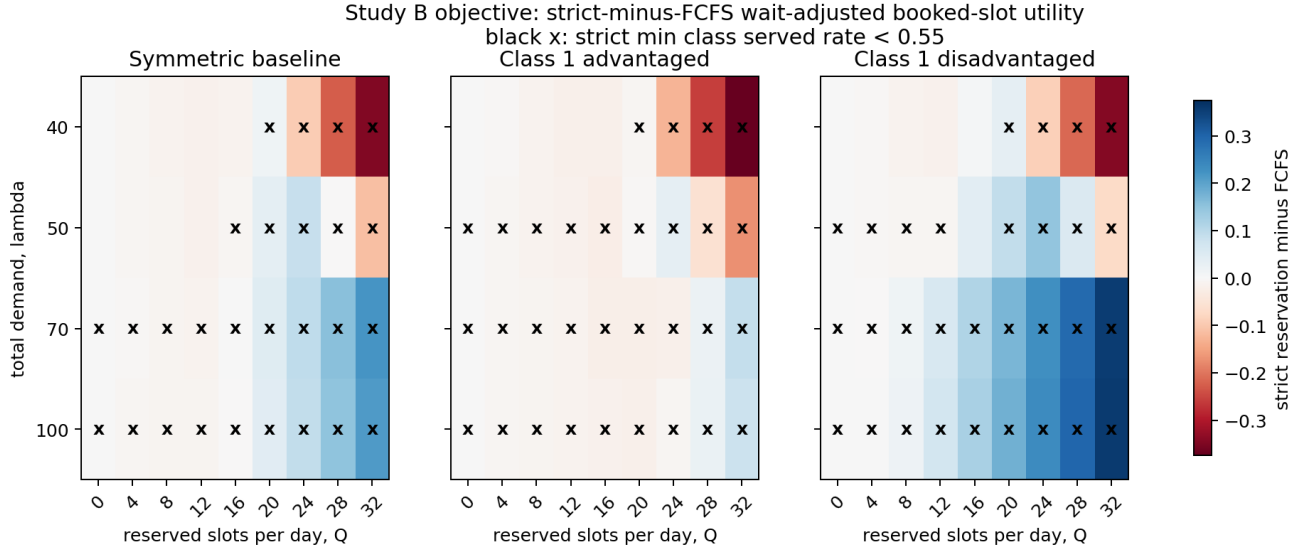


Figure 3: Study B objective: strict-minus-FCFS wait-adjusted booked-slot utility. Black x marks cells where strict reservation violates the service-rate threshold.

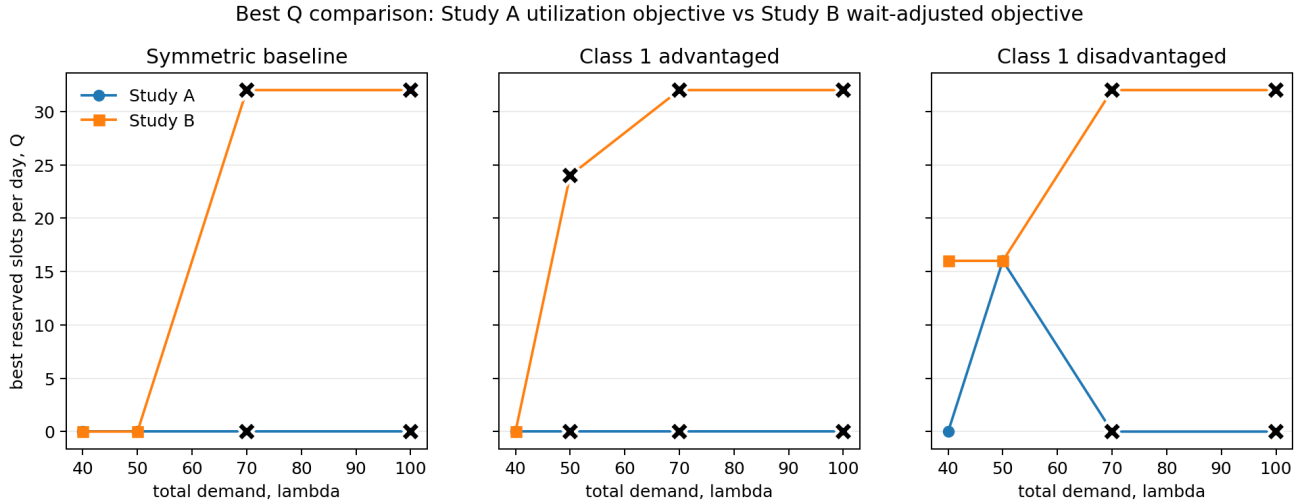


Figure 4: Best Q under each study. Black X markers with a white outline mark infeasible selected values.

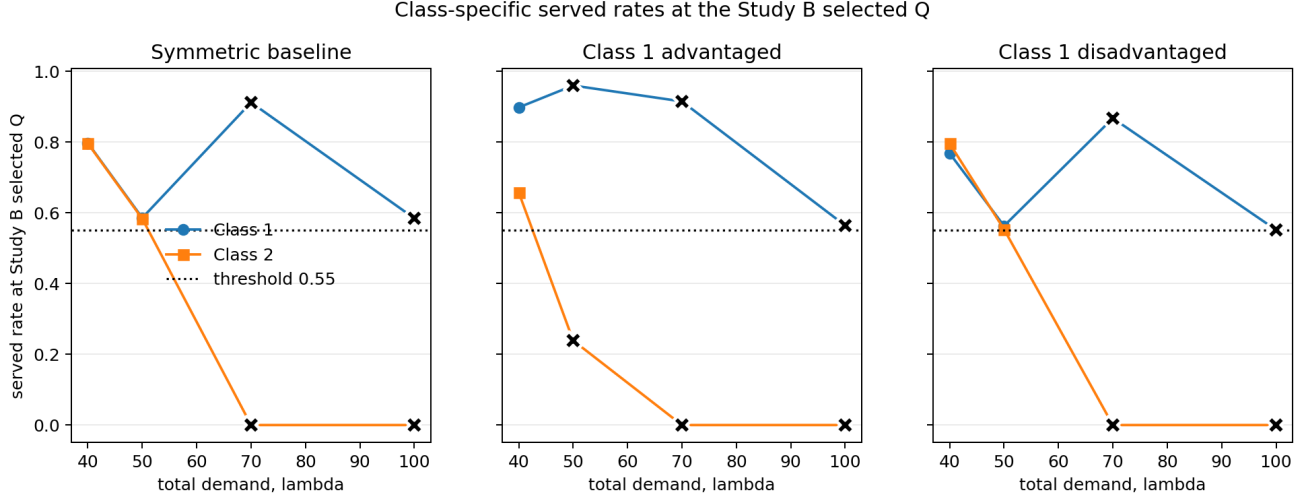


Figure 5: Class-specific served rates at the Study B selected Q . Class 1 and Class 2 are shown directly on the same axes; no class-advantage difference is plotted.

5 Sensitivity

The sensitivity table below focuses on the baseline setting and the headline Class 1 weight $w_1 = 1.5$. The main pattern is clear: without a protected-slot cost ($c = 0$), the wait-adjusted score can select a positive Q . Once $c = 0.05$ or $c = 0.10$ is used, the selected baseline Q returns to 0.

Study	w_1	c	γ	SR floor	Best Q	Beats FCFS	Failure mode
B	1.5	0.00	0.00	0.55	12	yes	none
B	1.5	0.00	0.02	0.55	12	yes	none
B	1.5	0.00	0.05	0.55	12	yes	none
B	1.5	0.00	0.10	0.55	12	yes	none
B	1.5	0.05	0.00	0.55	0	no	objective does not beat FCFS
B	1.5	0.05	0.02	0.55	0	no	objective does not beat FCFS
B	1.5	0.05	0.05	0.55	0	no	objective does not beat FCFS
B	1.5	0.05	0.10	0.55	0	no	objective does not beat FCFS
B	1.5	0.10	0.00	0.55	0	no	objective does not beat FCFS
B	1.5	0.10	0.02	0.55	0	no	objective does not beat FCFS
B	1.5	0.10	0.05	0.55	0	no	objective does not beat FCFS
B	1.5	0.10	0.10	0.55	0	no	objective does not beat FCFS

6 Decision Points Before Finalizing

- Confirm whether the service-rate floor should remain 0.55 or be raised.
- Decide whether booked-slot utilization is sufficient as the primary capacity target, or whether attended utilization should become a hard constraint because no-shows do not produce visits.
- Decide whether Class 2 served-rate loss should have its own explicit threshold.
- Rerun the sweep with more seeds before treating any Q recommendation as final.

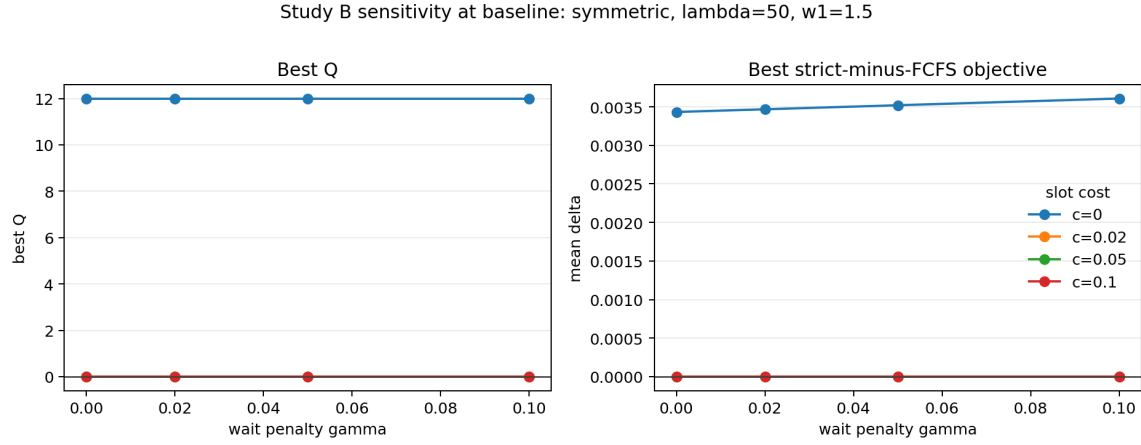


Figure 6: Study B sensitivity at the baseline scenario. The plotted objective is the best strict-minus-FCFS wait-adjusted utility for each waiting-time penalty and slot-cost value.

7 Audit Notes

- Earlier objective-function wording from the draft has not been reused.
- This memo analyzes only the strict Class 1 reservation policy against pooled FCFS.
- The metric `booked_slot_utilization` is available in the simulation output.
- Class offered-delay totals and class service-rate guardrails are available.
- Each study uses one main objective for best- Q selection.